JIAZHOU 'JOE' LIU

Lecturer

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liujiazhou@gmail.com

Personal Website

RESEARCH AND TEACHING INTERESTS

I am a Lecturer in the Embodied Visualisation (EmVis) Research Group at Monash University and an early-career researcher in Human–Computer Interaction (HCI). My research is situated at the intersection of Immersive Analytics, spatial cognition, and applied artificial intelligence (AI).

My research focuses on the design and evaluation of novel interaction techniques for immersive environments, with particular emphasis on human-in-the-loop systems and the integration of AI methods into immersive analytics. I lead and collaborate on a range of research projects that apply these approaches across multiple domains, including digital health, simulation-based training and education, and smart manufacturing.

I have over seven years of teaching experience at Monash University and the University of Melbourne, serving in roles as Chief Examiner, Unit Coordinator, Lecturer, and Tutor. I have taught undergraduate and postgraduate units in Research Methods, Data Visualisation, User Interface Design and Usability, Web Fundamentals, Mobile Application Development, and Full-Stack Development. I have also supervised PhD students, Master's minor theses, and final-year undergraduate research projects.

EDUCATION

Monash University
Doctor of Philosophy, Computer Science
2019 – 2023

Monash University
Master of Information Technology
2017 – 2018

University of New South Wales
Bachelor of Science, Computer Science
2010 - 2013

RESEARCH INTERESTS

- Human-Computer Interaction
- Data Visualisation
- Immersive Analytics
- Spatial Cognition and Skills
- Embodied Interaction
- Digital Health
- Construction Safety Management

RESEARCH PAPERS (SELECTED)

Jiazhou Liu, Kadek Satriadi, Barrett Ens, and Tim Dwyer, ***Investigating the Effects of Physical Landmarks on Spatial Memory for Information Visualisation in Augmented Reality***. 2024 IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2024)

Jiazhou Liu, Barrett Ens, Arnaud Prouzeau, Jim Smiley, Isobel Nixon, Sarah Goodwin, and Tim Dwyer, ***DataDancing: An Exploration of the Design Space for Visualisation View Management for 3D Surfaces and Spaces***. 2023 ACM CHI Conference on Human Factors in Computing Systems (ACM CHI 2023)

Ang Li, Jiazhou Liu, Maxime Cordeil, Jack Topliss, Thammathip Piumsomboon, and Barrett Ens, ***GestureExplorer: Immersive Visualisation and Exploration of Gesture Data***. 2023 ACM CHI Conference on Human Factors in Computing Systems (ACM CHI 2023)

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GRANTS, AWARDS & SCHOLARSHIPS

ECA Seed Grant
Monash Faculty of Information Technology
2024 - 2025 and 2025 - 2026

Postgraduate Publication Award
Monash Graduate Research
2023

Graduate Research Completion Award
Monash Graduate Research
2022

Australian Research Training Program Scholarship
Australian Department of Education
2019 - 2022

Information Technology Postgraduate Scholarship
Monash University
2018 - 2019

ACADEMIC SERVICES

Conference Committee
iiWAS/MoMM 2025 - PC Member
ACM ISS 2024 - Web Chair
IEEE ISMAR 2023 - Web Chair

Paper Reviewer
ACM CHI 2020 (LBW), 2023, 2024, 2025, 2026
IEEE ISMAR 2025
IEEE VR 2021, 2025, 2026
IEEE VIS 2024, PacificVis 2026
ACM ISS 2021, 2022
TVCG 2026

Workshop Organiser
IEEE VIS 2025
ACM CHI 2022
Monash Take CRTL 2021

RESEARCH PAPERS (CONT.)

Jiazhou Liu, Arnaud Prouzeau, Barrett Ens, and Tim Dwyer, ***Effects of Display Layout on Spatial Memory for Immersive Environments***. 2022 ACM Interactive Surfaces and Spaces Conference (ACM ISS 2022)

Rhys Newbury, Kadek Ananta Satriadi, Jesse Bolton, Jiazhou Liu, Maxime Cordeil, Arnaud Prouzeau, and Bernhard Jenny. ***Embodied Gesture Interaction for Immersive Maps***. 2021 Cartography and Geographic Information Science (CGIS '21)

Jiazhou Liu, Arnaud Prouzeau, Barrett Ens, and Tim Dwyer. ***Design and Evaluation of Interactive Small Multiples Data Visualisation in Immersive Spaces***. 2020 IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR '20) **[Best Paper Nomination]**

DEMONSTRATION, POSTERS, AND WORKSHOP PAPERS

Mengxing Li, Phoebe Zhang, Jiazhou Liu, et al. **HandovAR: Towards AR and AI Support for ICU Nurse Handover.** 2026 ACM CHI Conference on Human Factors in Computing Systems (ACM CHI 2026) | XR for Challenging Environments Workshop

Vahid Pooryousef, Himashi Peiris, Huiyi Liang, Ang Li, Zhaolin Chen, Tim Dwyer, Jiazhou Liu. **SAMMed-VR: Integrated Segment Anything Model in Virtual Reality for Supervised Brain Tumour Segmentation.** 2025 IEEE Visualisation (IEEE VIS 2025) Poster

Jiazhou Liu, Aravinda S. Rao, Fucai Ke, Tim Dwyer, Benjamin Tag, Pari Delir Haghighi. ***AR-Facilitated Safety Inspection and Fall Hazard Detection on Construction Sites.*** 2024 IEEE International Symposium on Mixed and Augmented Reality.

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TALKS & PRESENTATIONS

Conference Presentation

IEEE ISMAR 2024
ACM CHI 2023
MEL-CHI 2023
ACM ISS 2022
IEEE VR 2020

Workshop Presentation

IEEE ISMAR 2024 - SafeAR
Monash Take CTRL 2021

Guest Lecturer/Presentation

Monash Children's Hospital 2025
Monash FIT3179 Data Visualisation 2021, 2022

TECHNICAL SKILLS

Proficient

Kotlin, Java, Swift, Swift UI, C#, Unity, Tableau, R, Vega-lite, JavaScript, D3, HTML, CSS, Adobe software, Python, AngularJS, NodeJS, SQL, Git

Familiar

C, SharePoint, VUE, Flutter

REFERENCES

Prof. Tim Dwyer

Performance Supervisor - Monash
Email: tim.dwyer@monash.edu

A/Prof. Bernhard Jenny

Teaching Unit Chief Examiner - Monash
Email: Bernie.Jenny@monash.edu

Dr Pari Delir Haghighi

Industry-funded Project Leader - Monash
Email: Pari.Delir.Haghighi@monash.edu

TEACHING ACTIVITIES

University of Melbourne

- COMP90044 Research Methods (**Tutor**) [2024]

Monash University

- FIT2095 Full stack development (**Chief Examiner**, **Lecturer**) [2024 - 2026]
- FIT2081 Mobile Application Development **(Lecturer)** [2024 - 2026]
- FIT1050 Web Fundamentals (**Lecturer**) [2024]
- FIT3179 Data Visualisation **(Lecturer/Head Tutor)** [2020 - 2025]
- FIT316X Computer Science Project **(Supervisor)** [2019, 2021 - 2022]
- FIT5147 Data Exploration and Visualisation **(Tutor)** [2022]
- FIT3175 Usability **(Tutor)** [2022]
- FIT5152 User Interface Design and Usability **(Tutor)** [2021]

INDUSTRY EXPERIENCE

IT Support Subject Matter Expert

Monash University, Clayton, VIC
2021 – 2023

SharePoint Consultant/Front-end Developer (Full-time)

WebVine, Redfern, NSW
2015 - 2016

Website Developer & Administrator (Full-time)

HBC International, Auburn, NSW
2015

Network Technician (Full-time)

HBC International, Auburn, NSW
2014 - 2015